در صورتی که دستورات ما در تکرارنیست از Ad-Hoc استفاده می‌کنیم مثل ست کردن یک پارامتر

Ansible Ad-Hoc command

مسیر اجرای دستورات Ansible

- ✓An Ansible ad-hoc command uses the */usr/bin/ansible* command-line tool to automate a single task on one or more managed nodes.
- ✓Ad-hoc commands are great for tasks you repeat rarely and they are not reusable.

```
# ansible [pattern] -m [module] -a "[module options]" host
```




Lets run some ad-hoc commands!

هر کدام از ماژول های Ansible که فایل یا بیتی ست آن قرار دارد که یک ورودی از ما میخواهد

ansible –m MODULE –a COMMAND HOSTS

Examples:

ansible –m ping 172.20.30.40

ansible –m shell –a 'ls /home' 172.20.30.40

ماژول shell مثل command که قابل اجرا باشد در shell متعدد

کجا اجرا شود؟

ماژول — ورودی

ansible –m shell –a 'df -h' 172.20.30.40

ansible –m shell –a 'whoami' 172.20.30.40

ansible –m setup 172.20.30.40 ⟶ All Host Info

ansible –m setup –a 'filter=ansible_user_id' 172.20.30.40

ansible -m shell -a "cat /etc/os-release" myservers



ansible myservers -m shell -a "df -h && free -m && hostname && ls /home"
ansible myservers -m setup -a "filter=ansible_os_family"

Exercise 2: run ansible ad-hoc command

1. Setup two another VMs.
2. Copy the ssh-key from the ansible server to your new VMs
3. Add the ip of the servers into /etc/ansible/hosts as a group named "myservers";
4. **Run** the following command **on both servers** via ansible **ad-hoc** command
    1. df -h
    2. free -m
    3. hostname
    4. ls /home
    5. -m setup -a "filter=ansible_os_family"

فرض کنید که یک گروه تعریف کردیم شامل ۱۰۰ تا host میخواهیم دستور فقط روی یک host

limit the command to a specific host

انجام شود limit ای گذاریم

You can use the --limit argument to limit the command to a specific host in the specified group. Assume the *myservers* group includes two servers (192.168.43.74 and 192.168.43.174). But you want to run a command only on 43.74:

ansible myservers -a "free -m" --limit 192.168.43.74

```
192.168.43.74 | CHANGED | rc=0 >>
              total        used        free      shared  buff/cache   available
Mem:            981         156         412           1         412         671
Swap:          1705           0        1705
```

ansible myservers --limit 192.168.40.137 -m shell -a "ip -br a"

Run commands with sudo

The **-b** option (alias for --become) tells Ansible to run the command with become (basically, run commands with 'sudo').

you should also pass in **-K** (alias for --ask-become-pass), so you can enter your password when Ansible needs it.

ansible -b -m yum -a "name=ntp state=present" myservers -K

BECOME password:

برای اجرای دستورات با دسترسی sudo این 2 سوییچ بالا میزنن

b- یعنی با دسترسی root اجرا کن (Become)

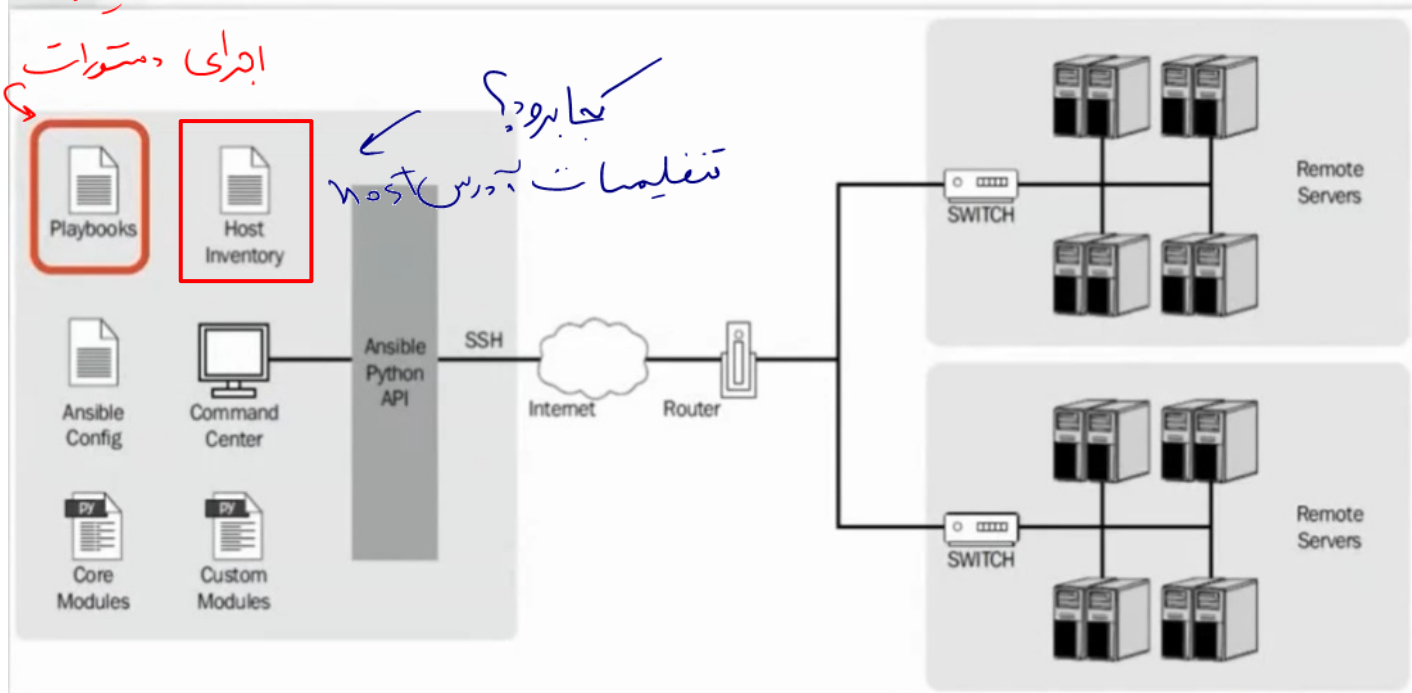
K- یعنی رمز عبور root رو ازم بپرس (Ask Pass)

Ansible Architecture ...

حالا میگه کجا؟

اجرای دستورات

کجا برود؟ تنظیمات آدرس Host

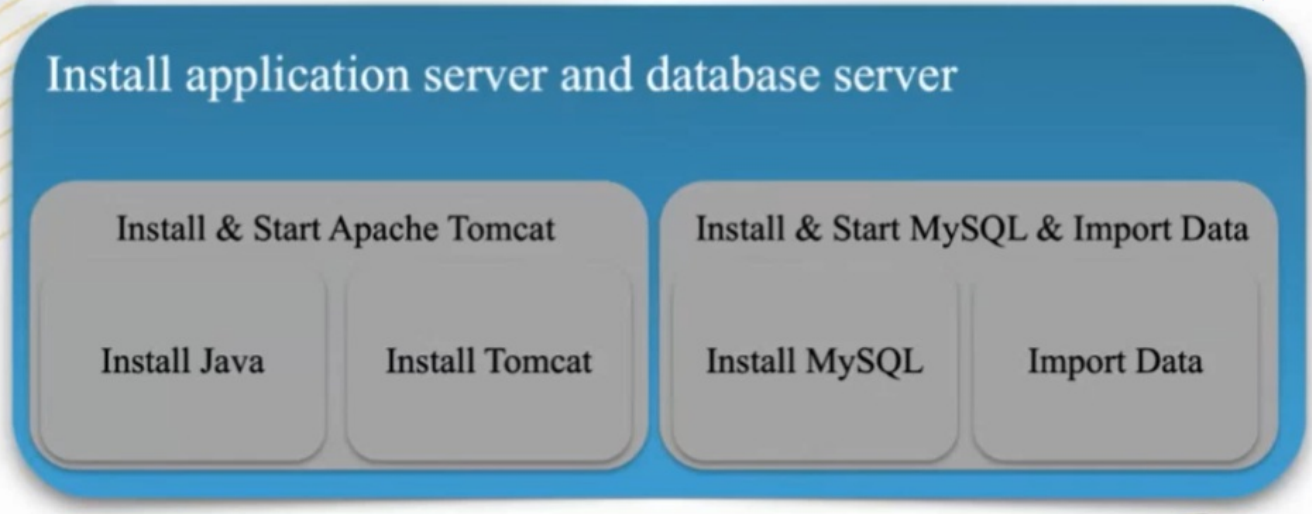


مثل یک to-do لیست هست که هر task رو انجام می‌دهد یک تایپی زنه

Ansible Architecture ...

PlayBooks tell Ansible what to execute and are like a to-do list for Ansible that contains a list of plays.

PlayBooks ها شامل یک list از Play ها هستند.



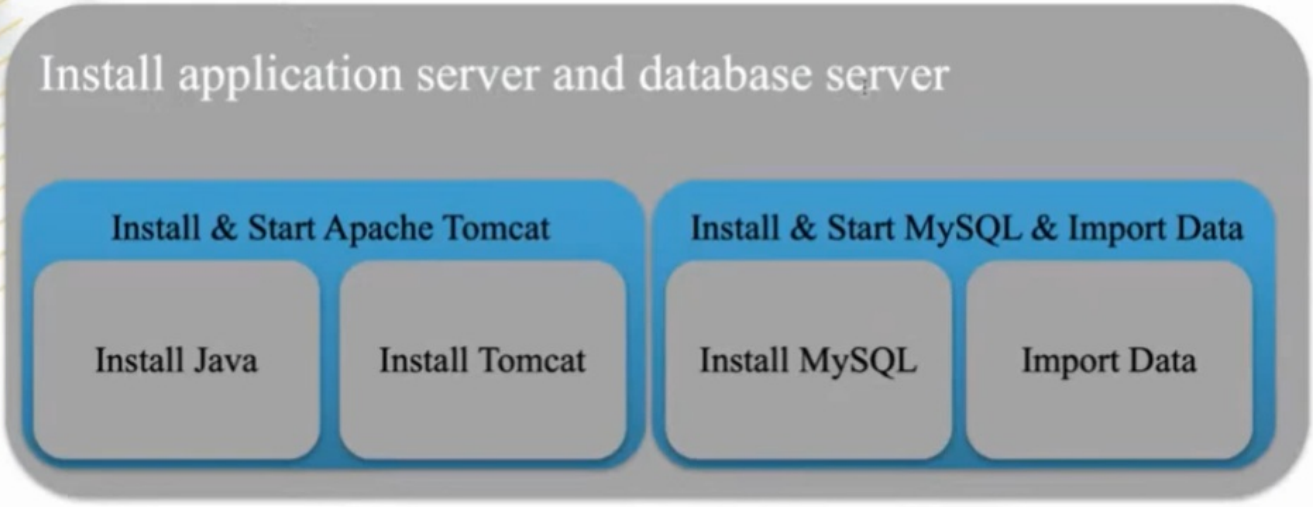


رول و نقش remot ماشین های ما را مشخص می‌کنند

Ansible Architecture ...

به مجموعه‌ای از این Play ها کنار هم یک PlayBook رو تشکیل می‌دهند.

Plays define the role that host machines will perform.

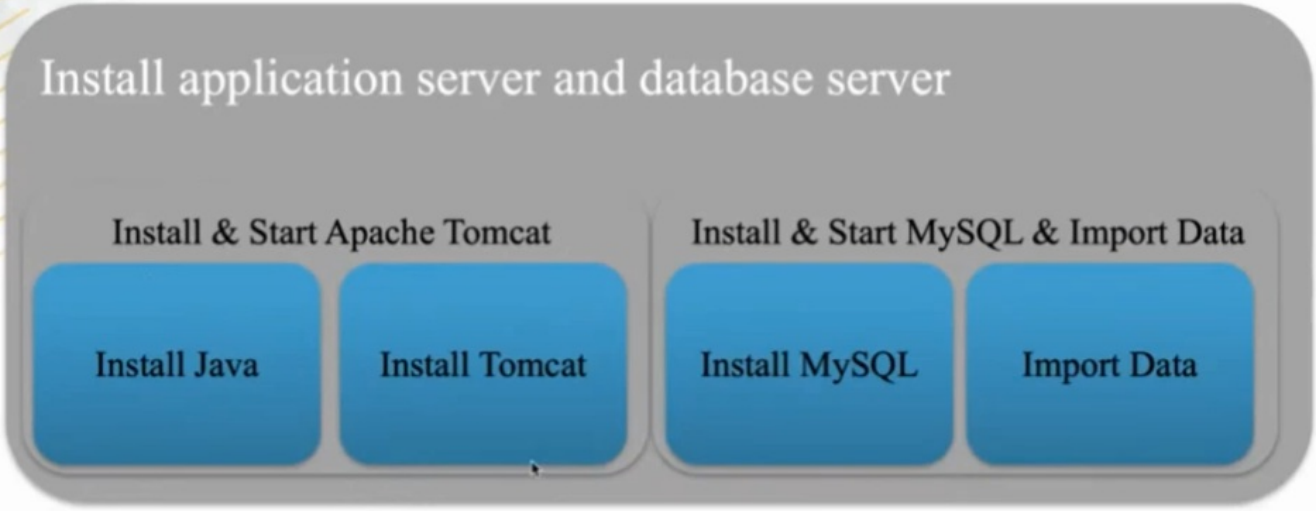




Ansible Architecture ...

Tasks combine an action with a name and optionally some other keywords. Tasks also call modules.

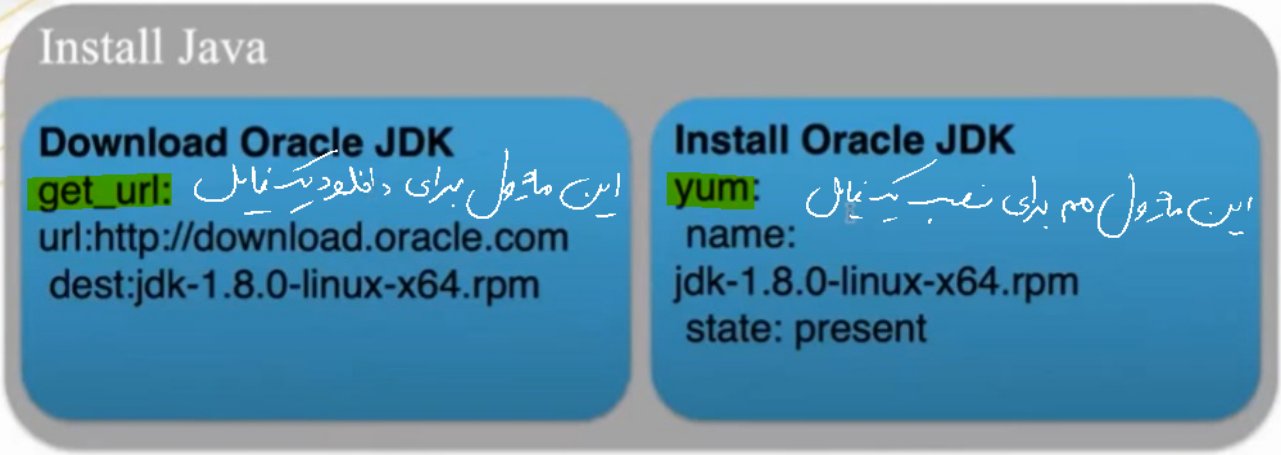
هر Play از مجموعه‌ای task ها ایجاد می‌شود



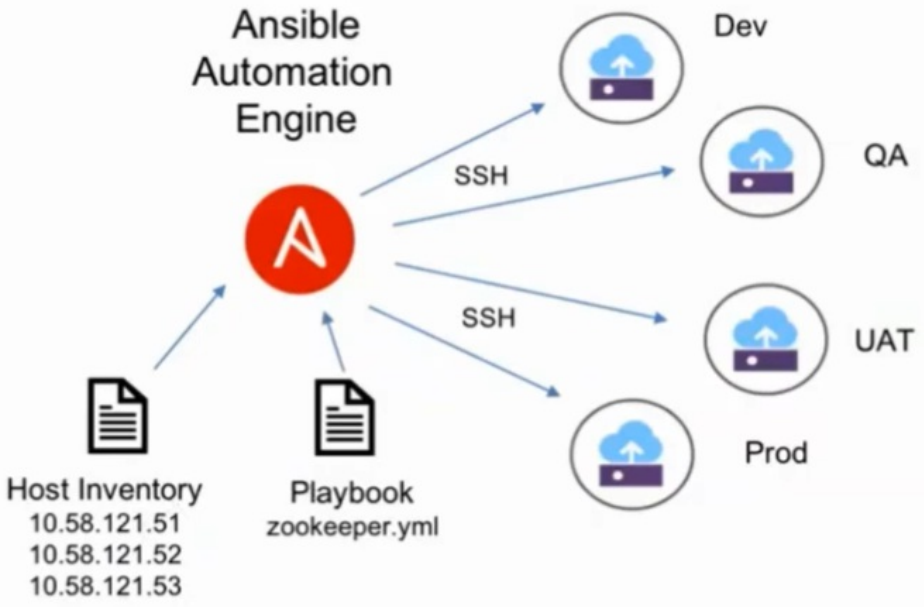
Ansible Architecture ...

Modules are the units of work that Ansible ships out to remote machines. They can be written in any language with the condition that its output should be in the JSON format.

هر task هم از یه سری Modules تشکیل گرفته است

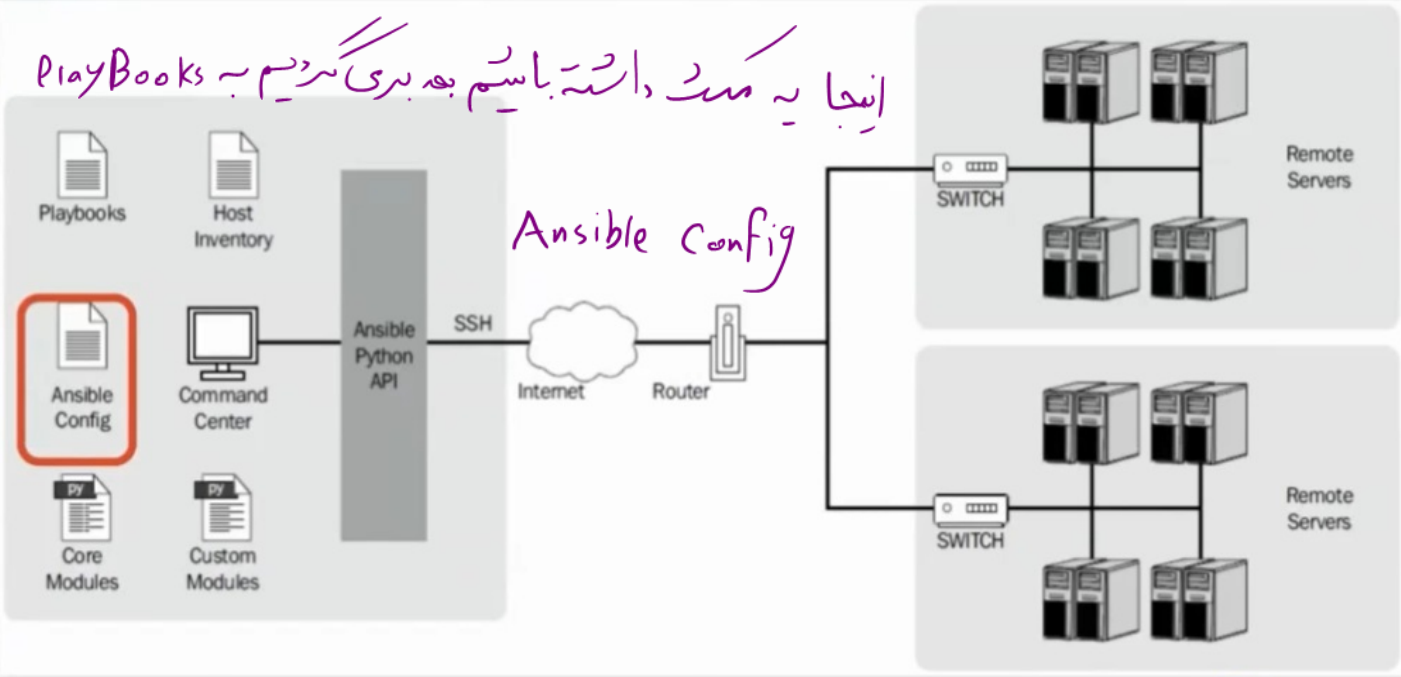


Ansible Automation Engine



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Ansible Architecture ...



Ansible Configuration ...

Ansible looks for its **configuration file** in a predefined order:

1. **ANSIBLE_CONFIG**
2. **./ansible.cfg**
3. **~/.ansible.cfg**
4. **/etc/ansible/ansible.cfg**

Ansible Configuration ...

- **Configuration using environment variables**

Use configuration parameters directly via environment variables by appending **ANSIBLE_** to the parameter in uppercase.

- **Example:**

```
export ANSIBLE_SUDO_USER=root
```




Ansible Configuration ...

- **Configuration using ansible.cfg**

host inventory =

- **Inventory:** the path to the inventory file

```
Inventory = /etc/ansible/hosts
```

- **library:** the path of Ansible modules

برای معرفی و اضافه کردن Coustome modules

```
Library = /usr/share/my_modules/
```

- **forks:** maximum processes in parallel

مشخص می‌کند که بصورت Parallel روی متامایشن چه تعداد کار موازی انجام بدهد

```
forks = 5
```

- **remote_port:** the port used for SSH connections

```
remote_port = 22
```

پورتی هست که ما برای ssh زدن به host ها استفاده می‌شود

پیش فرض روی 22 هست



Ansible Configuration ...

- **Configuration using ansible.cfg**
  - **Log_path:** the path of Ansible logs

```
log_path = /var/log/ansible.log
```

  - **transfer_method:** the mechanism for transferring files

```
transfer_method = smart/sftp/scp/piped
```

  - **connect_timeout:** connection timeout value in seconds

```
connect_timeout = 30
```




Ansible Configuration ...

- **Configuration using ansible.cfg**
  - **remote_tmp:** transferring modules to your remote machines, running them, and then cleaning up after itself

```
Remote_tmp = ~/.ansible/tmp
```

  - **Roles_path:** additional paths to search for roles in, colon separated

```
Roles_path = /etc/ansible/roles
```




Ansible Configuration ...

- **Configuration using ansible.cfg**
  - **ask_pass:** This controls whether an Ansible playbook should prompt for a password by default. The default behavior is no:

```
ask_pass = True
```

≠ false

Prompt رو بعدت Pass بیرد

  - **ask_sudo_pass:** Similar to ask_pass, this controls whether an Ansible playbook should prompt for a sudo password by default when sudoing. The default behavior is also no:

```
ask_sudo_pass = True
```

← حالا اگه sudo Pass بود

  - **ask_vault_pass:** This controls whether an Ansible playbook should prompt for the vault password by default. The default behavior is no:

```
ask_vault_pass = True
```




Ansible Configuration ...

- **Configuration using ansible.cfg**

3 دسته warning داریم

  - **command_warnings:** By default since Ansible 1.8, Ansible will issue a warning when the shell or command module is used and the command appears to be similar to an existing Ansible module.

به شما توصیه می کنه از ماژول که بهتره استفاده کنی

```
command_warnings = False
```

  - **deprecation_warnings:** Allows disabling of deprecating warnings in ansible-playbook output:

```
deprecation_warnings = True
```

یه سری از ماژول های که استناده میکنیم ممکن هست نسخه آپدیت های بعدی دیگه تخمیر شود که هشدار بده

  - **system_warnings:** Ansible may display warnings based on the configuration of the system running ansible itself. This may include warnings about 3rd party packages or other conditions that should be resolved if possible.

هشدارهای سیستمی

```
system_warnings = True
```




Ansible Configuration ...

- **Configuration using ansible.cfg**

setup ⇒ جمع‌آوری اطلاعات

Gathering: the 'gathering' setting controls the default policy of facts gathering (variables discovered about remote systems).

یک بار اطلاعات روی گیرد و سری بعدی از cash می‌گیرد

smart - gather by default, but don't regather if already gathered

implicit - gather by default, turn off with gather_facts: False

بصورت پیش فرض اطلاعات رو بگیر و اگر ما خواهیم باید False کنیم

explicit - do not gather by default, must say gather_facts: True

برعکس بالا

gathering = implicit



Ansible Configuration ...

- **Configuration using ansible.cfg**
  - **Gathering:** You can specify a subset of gathered facts, via the play's gather_facts directive, using the following option:

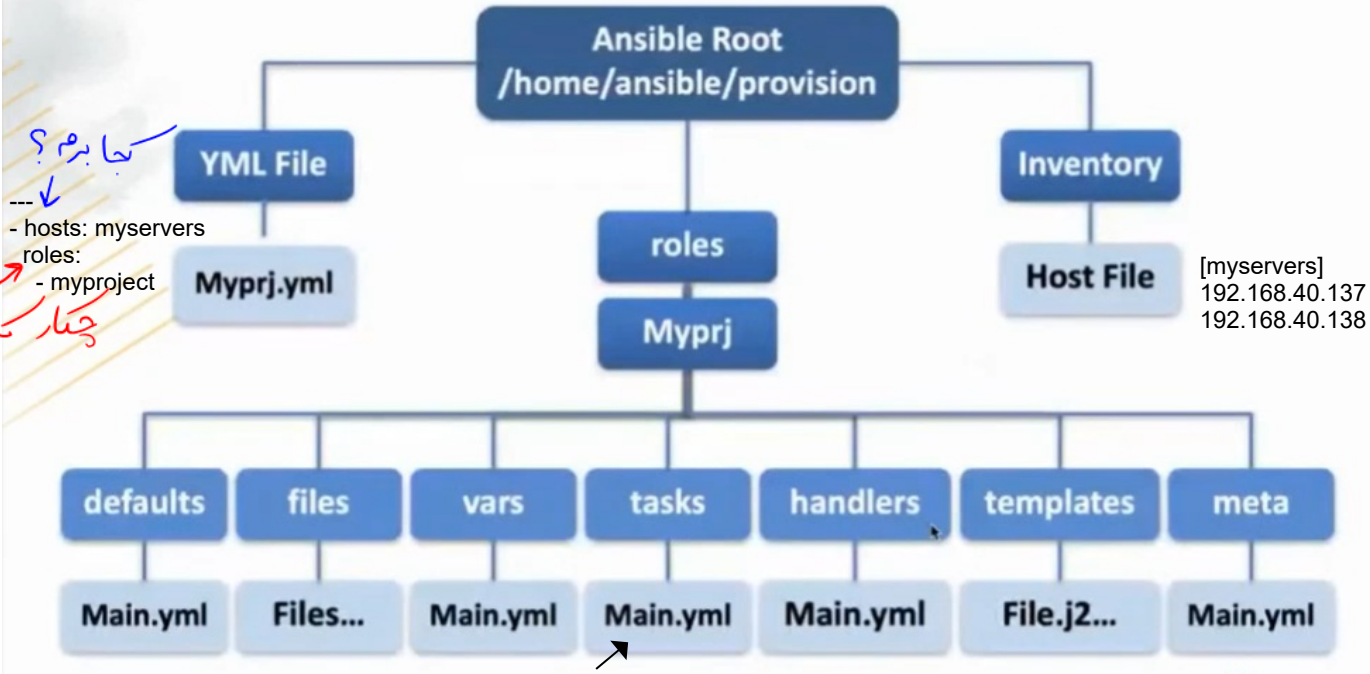
all:	gather all subsets (the default)
network:	gather network facts
hardware:	gather hardware facts (longest facts to retrieve)
virtual:	gather facts about virtual machines hosted on the machine
ohai:	gather facts from ohai
facter:	gather facts from facter

gather_subset = all

اگه همه دیتا رو خواهیم



Ansible-playbook Structure



- hosts: myservers
  roles:
    - myproject

[myservers]
192.168.40.137
192.168.40.138

```
- name: "******** Create Dir1 In /home *********"
  file:
    path: /home/dir1
    state: directory
    owner: root
    group: root
    mode: 755
  tags: create_dir1
```




Ansible-playbook Structure

```
# mkdir –p /home/ansible/provision

# vim /home/ansible/provision/myproject.yml

# mkdir /home/ansible/provision/inventory

# vim /home/ansible/provision/inventory/Hosts

# mkdir /home/ansible/provision/roles

# mkdir /home/ansible/provision/roles/myproject
```

- files
- vars
- defaults
- templates
- handlers
- tasks
- meta



Exercise 3: Setup an Ansible-playbook structure

1. Create the following structure in /home/ansible/provision

Ansible root

```
.
├── inventory
│   └── anisahosts
├── myproject.yml
└── roles
    └── myproject
        ├── defaults
        │   └── main.yml
        ├── files
        ├── handlers
        │   └── main.yml
        ├── meta
        │   └── main.yml
        ├── tasks
        │   └── main.yml
        ├── templates
        └── vars
            └── main.yml
```




بسته بندی و سازماندهی کدهای Ansible در یک ساختار فایل مشخص

Ansible-playbook Structure

مثل یک template میمونه

- ✓ **Roles** let you automatically load related vars_files, tasks, handlers, and other Ansible artifacts based on a known file structure. Once you group your content in roles, you can easily reuse them and share them with other users.
- **Tasks:** A list of actions you want to perform on the remote machines and is a declaration about the state of them.

همون to-do هاست که کارهای که گفتیم میاد مرحله به مرحله انجام میده

  - **Examples:**
    - Directory should exist
    - Package should be installed
    - Service should be running
    - Configuration file should be changed



Ansible-playbook Structure

- **vars:** A list of variables which will be used by the playbook
- **files:** A list of source files (binaries, rpms, ...)
- **templates:** A list of configuration files with jinja2 extension
- **handlers:** handles the state of services
- **meta:** metadata, prerequisite actions and dependencies



Ansible-playbook Syntax

Main YML File:

vim /home/ansible/provision/ProjectName.yml

```
---
- hosts: test-hosts        #(List of Hosts in Inventory Directory)
  roles:                   # (roles Directory )
    - myproject            # (Project Directory in roles)
```